Machine Learning Lectures

Regression

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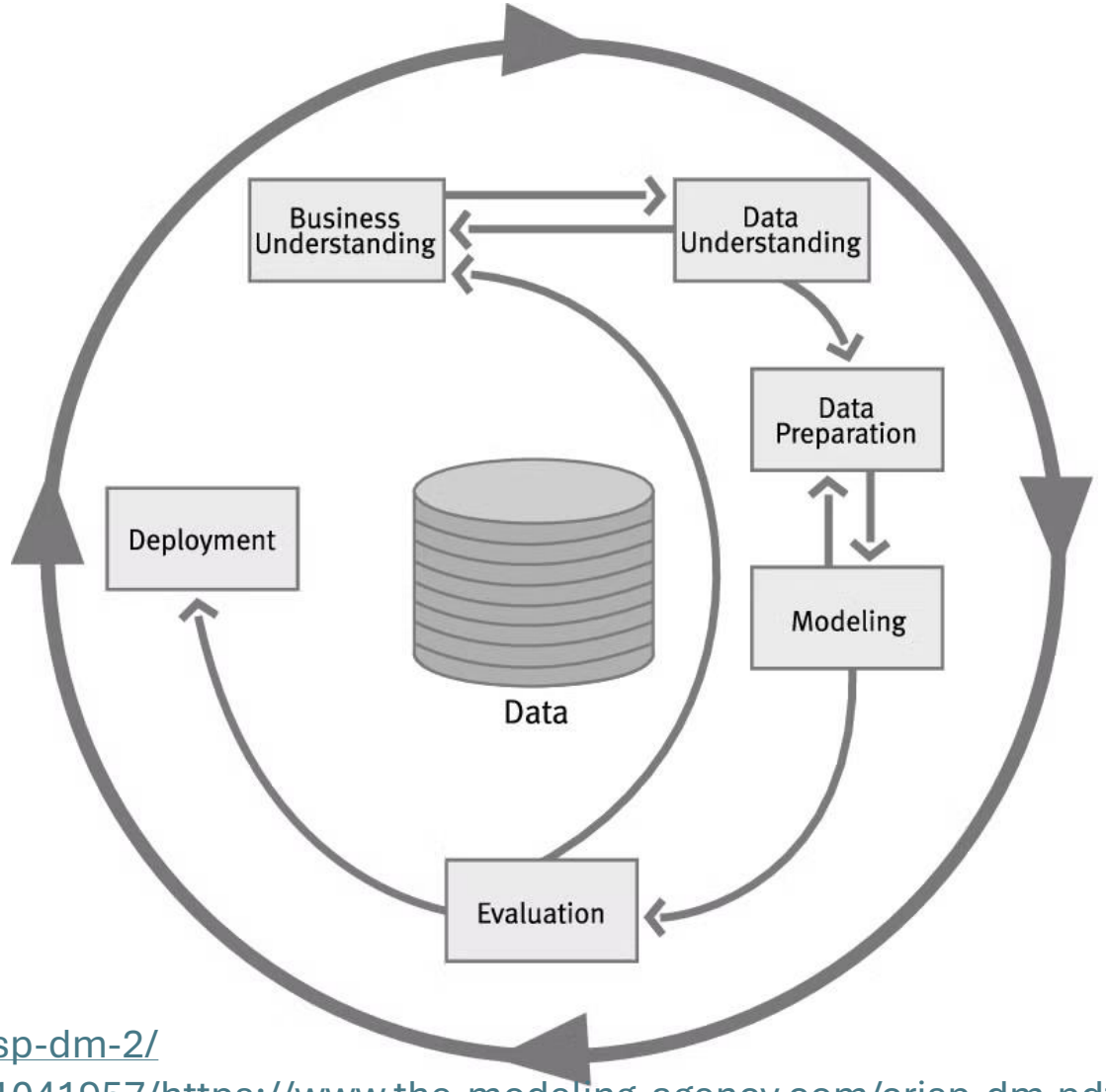
Agenda

1. Regression Problem Setting
2. Basic Regression Models and Data Preparation
3. Multicollinearity Problem
4. Model Evaluation and Validation
5. Practical Considerations
6. Advanced Topics in Regression (Optional)

Data Analytics pipeline: CRISP-DM



https://www.datascience-pm.com/crisp-dm-2/
https://web.archive.org/web/20220401041957/https://www.the-modeling-agency.com/crisp-dm.pdf

Business Understanding

Determine Business Objectives
Background
Business Objectives
Business Success Criteria

Assess Situation
Inventory of Resources
Requirements, Assumptions, and Constraints
Risks and Contingencies
Terminology
Costs and Benefits

Determine Data Mining Goals
Data Mining Goals
Data Mining Success Criteria

Produce Project Plan
Project Plan
Initial Assessment of Tools and Techniques

Data Understanding

Collect Initial Data
Initial Data Collection Report

Describe Data
Data Description Report

Explore Data
Data Exploration Report

Verify Data Quality
Data Quality Report

Data Preparation

Select Data
Rationale for Inclusion/ Exclusion

Clean Data
Data Cleaning Report

Construct Data
Derived Attributes
Generated Records

Integrate Data
Merged Data

Format Data
Reformatted Data

Dataset
Dataset Description

Modeling

Select Modeling Techniques
Modeling Technique
Modeling Assumptions

Generate Test Design
Test Design

Build Model
Parameter Settings
Models
Model Descriptions

Assess Model
Model Assessment
Revised Parameter Settings

Evaluation

Evaluate Results
Assessment of Data Mining Results w.r.t. Business Success Criteria
Approved Models

Review Process
Review of Process

Determine Next Steps
List of Possible Actions
Decision

Deployment

Plan Deployment
Deployment Plan

Plan Monitoring and Maintenance
Monitoring and Maintenance Plan

Produce Final Report
Final Report
Final Presentation

Review Project
Experience Documentation

Introduction to Regression

Regression Problem

Regression models the relationship between a dependent variable (**target**) and one or more independent variables (**features**)

Real-World Examples:

- Predicting house prices
- Stock market trends,
- Sales forecasting,
- Disease progression modeling.

Types of Regression:

- Simple Regression (one feature)
- Multiple Regression (many features)
- Polynomial Regression (non-linear relationships)

Regression Problem

Suppose we have a dataset of pairs:

$$\{(X_i, Y_i)\}_{i=1}^{m}$$

where:

$$X_i \in \mathbb{R}^n$$

represents the i -th observation of n-dimensional features, and

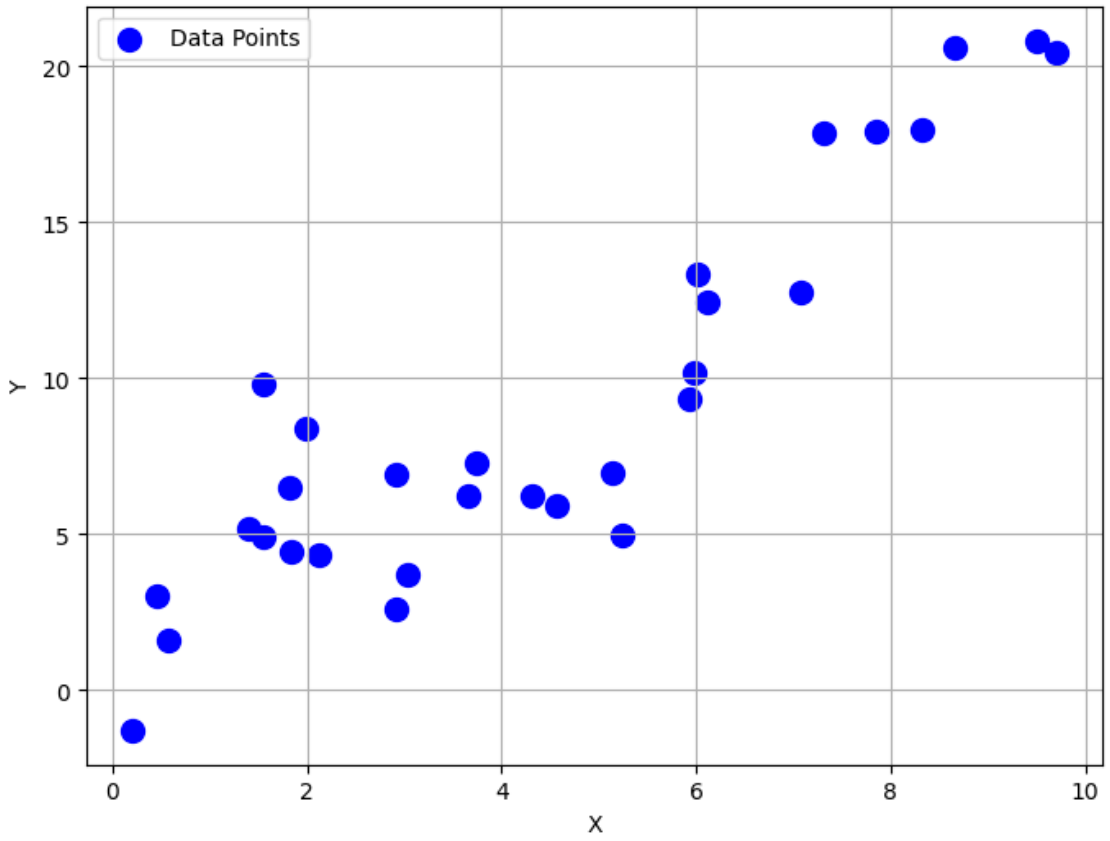
$$Y_i \in \mathbb{R}$$

is the corresponding continuous target value.

The goal is to approximate the unknown relationship:

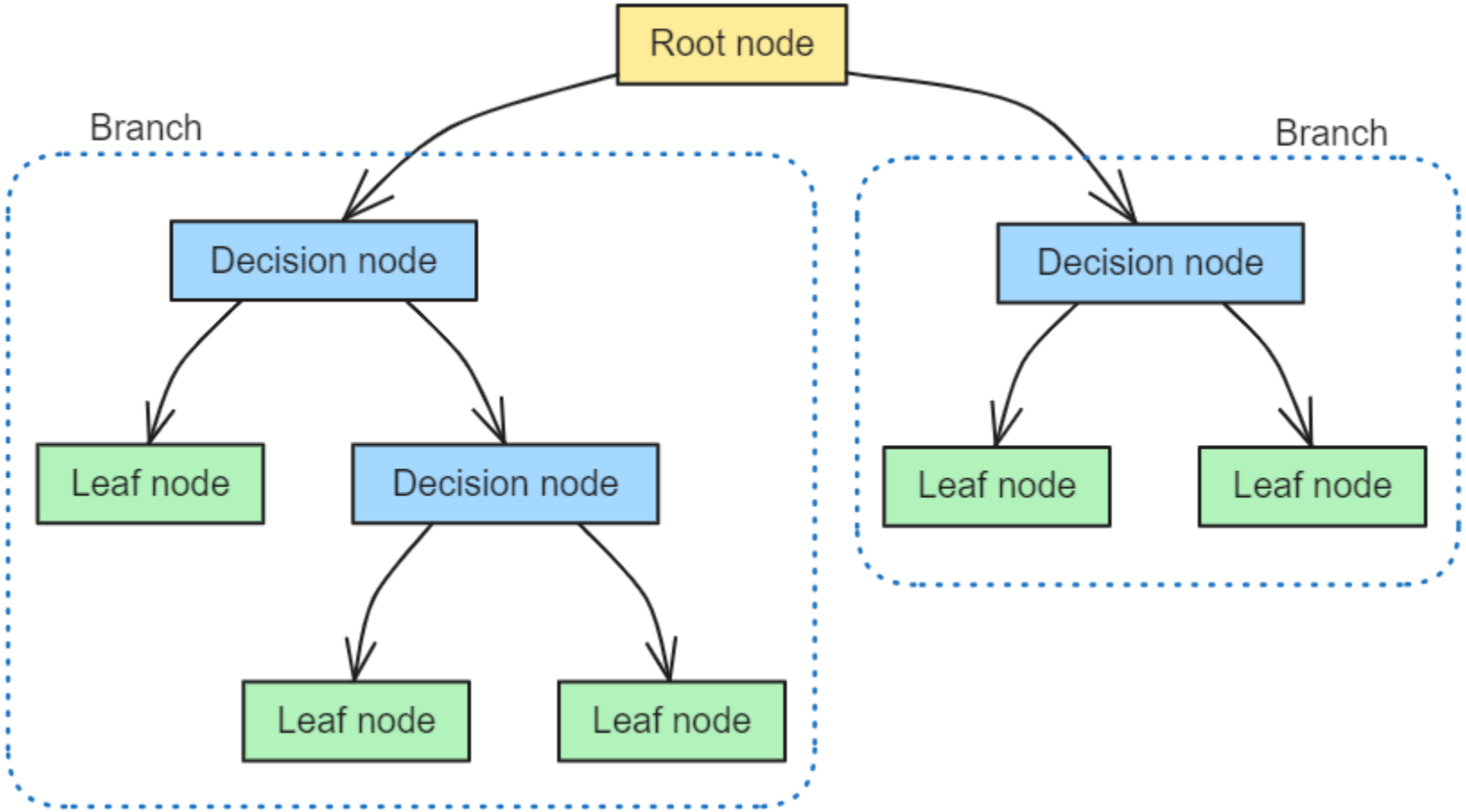
$$Y = f(X) + \epsilon$$

where ϵ represents the random noise or error term, usually assumed to have zero mean $\mathbb{E}[\epsilon] = 0$



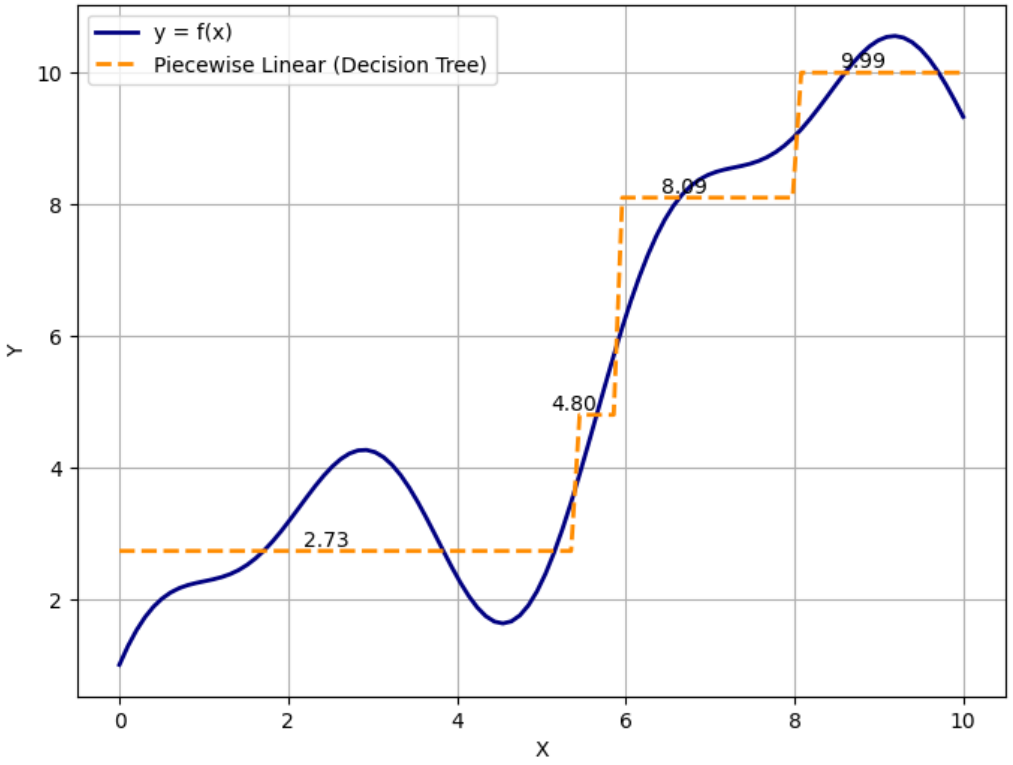
Basic Regression Models

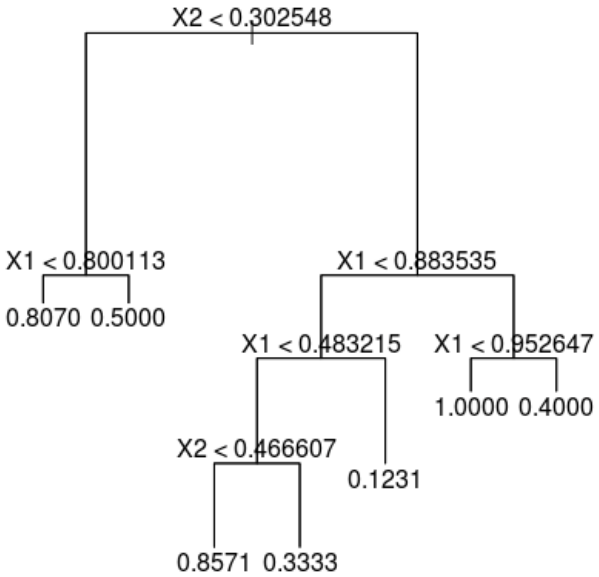
Decision Tree

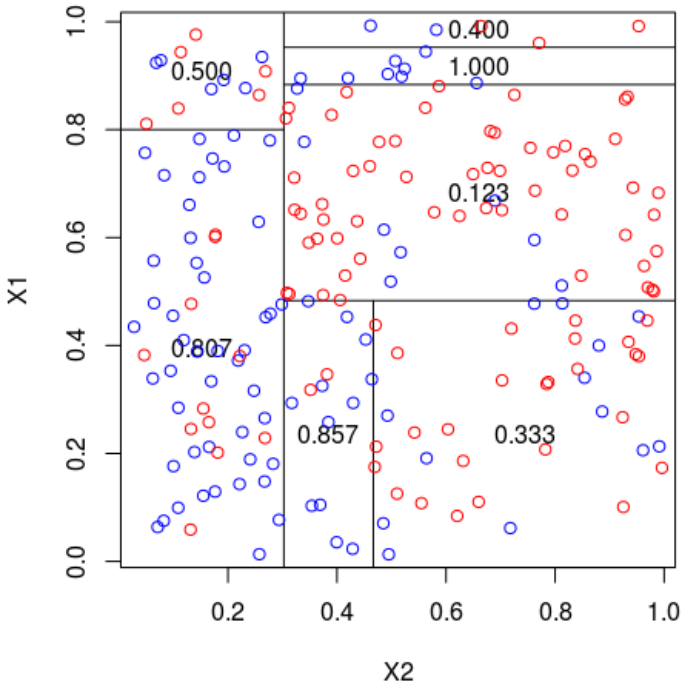


Video Explanation:
https://youtu.be/UhY5vPfQIrA?si=juxgJJHnso_Mj4R0

Decision Tree Regressor







Video Explanation:
https://youtu.be/UhY5vPfQIrA?si=juxgJJHnso_Mj4R0

Linear Regression

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i = \mathrm{x}_i^{\mathsf{T}} \beta + \epsilon_i \quad i = 1, \ldots, n.$$

$$\boxed{Y = X\beta + \epsilon}$$

$$Y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

a vector of **observed** values of the variable called the ***regressand***, ***endogenous variable***, ***response variable***, ***<u>target variable</u>***.

$$X = \begin{bmatrix} 1 & x_{11} & \ldots & x_{1p} \\ 1 & x_{21} & \ldots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \ldots & x_{np} \end{bmatrix}$$

X can be treated as either observed values.
X contains the **independent variables** (**regressors**) used in a regression model. Regressors can include a **constant (intercept)**, usually denoted as $x_{i0} = 1$, to account for a baseline value in the model.

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}$$

(p+1)-dimensional parameter vector, where β_0 is the intercept term (if one is included in the model, otherwise β is p-dimensional). Its elements are known as **effects** or **regression coefficients**

$$\epsilon = \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

the *error term*, ***disturbance term***, ***noise***. This variable captures all other factors which influence the dependent variable *y* other than the regressors **x**.

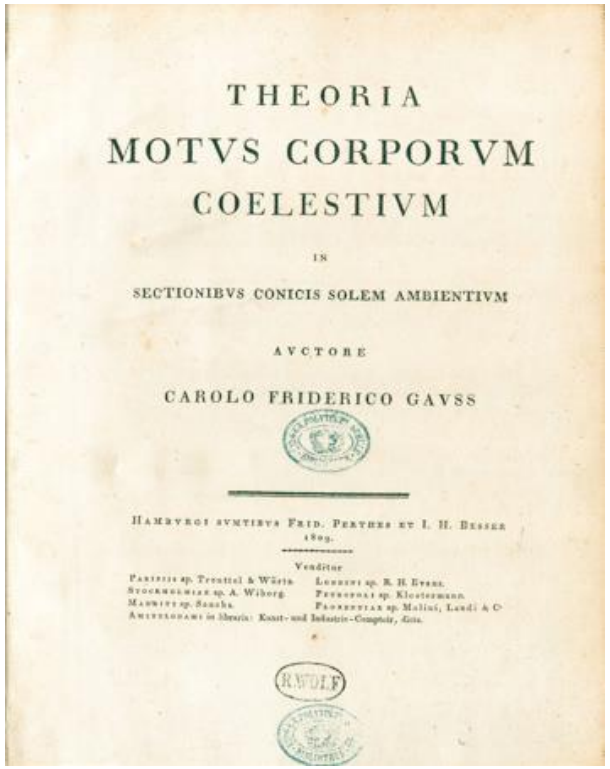
How to solve Linear Regression?

Ordinary Least Squares (OLS): A Historical overview

Gauss – (likely, not proven) invention, claimed to had been using it since **1794/95**

Carl Friedrich Gauss,
1777 - 1855

https://books.google.de/books?id=ORUOAAAAQAAJ



THEORIA
MOTVS CORPORVM
COELESTIVM
IN
SECTIONIBVS CONICIS SOLEM AMBIENTIVM
AVCTORE
CAROLO FRIDERICO GAVSS

HAMBVRGI SVMTIBVS FRID. PERTHES ET I. H. BESSER
1809.

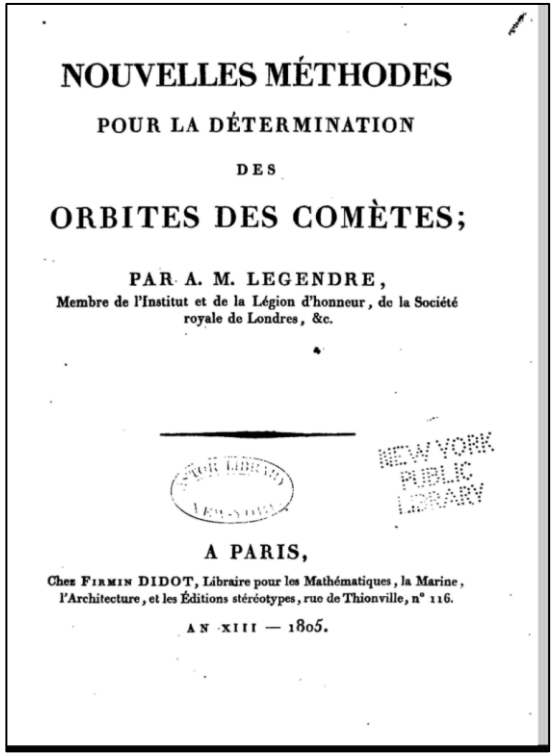
Theoria motus (1809)

Legendre – first published in **1805**

Nouvelles méthodes pour la détermination des orbites des comètes

Adrien-Marie Legendre
1752 - 1833

https://books.google.de/books?id=FRcOAAAAQAAJ&pg=PA1



NOUVELLES MÉTHODES
POUR LA DÉTERMINATION
DES
ORBITES DES COMÈTES;

PAR A. M. LEGENDRE,
Membre de l'Institut et de la Légion d'honneur, de la Société royale de Londres, &c.

A PARIS,
Chez FIRMIN DIDOT, Libraire pour les Mathématiques, la Marine, l'Architecture, et les Éditions stéréotypes, rue de Thionville, n° 116.
AN XIII — 1805.

First Definition

APPENDICE.

Sur la Méthode des moindres quarrés.

Dans la plupart des questions où il s'agit de tirer des mesures données par l'observation, les résultats les plus exacts qu'elles peuvent offrir, on est presque toujours conduit à un système d'équations de la forme

$$\boxed{E = a + bx + cy + fz + \&c.}$$

dans lesquelles a, b, c, f, &c. sont des coëfficiens connus, qui varient d'une équation à l'autre, et x, y, z, &c. sont des inconnues qu'il faut déterminer par la condition que la valeur de E se réduise, pour chaque équation, à une quantité ou nulle ou très-petite.

Ordinary Least Squares (OLS): Definition

n linear equations in p
unknown coefficients with n> p

$$\sum_{i=1}^{p} x_{ij}\,\beta_j = y_i \qquad i = 1, \dots, n.$$

(overdetermined system)

$$\boxed{y = X\beta + \epsilon}$$ ← Model

$$\boxed{\hat{y} = X\hat{\beta}}$$ ← Estimation

Goal: Estimate the coefficients $\beta_0, \beta_1, \dots, \beta_p$ such that the sum of squared residuals is minimized.

$$\hat{\beta} = \arg\min_{\beta} S(\beta) \qquad S(\beta) = (y - X\beta)^T\,(y - X\beta)$$

$S(\hat{\beta}) = RSS$ – Residual Sum of Squares (aka *SSR*)

Exact Solution of OLS

$$y = X\beta + \epsilon \qquad \hat{\beta} = \arg\min_{\beta} S(\beta) \qquad S(\beta) = (y - X\beta)^T (y - X\beta)$$

$$S(\beta) = y^T y - 2y^T X\beta + \beta^T X^T X\,\beta$$

$$\frac{dS(\beta)}{d\beta} = -2X^T y + 2X^T X\,\beta = 0$$

$$X^T X\,\beta = y^T X$$

$$\boxed{\hat{\beta} = (X^T X)^{-1} X^T y}$$

unbiased estimation of sigma

$$\boxed{\widehat{\sigma^2} = \frac{1}{n-p-1} e^T e}$$

$e^T e = SSR$ *n-p-1 = degrees of freedom.*
Let us skip the calculation for simplicity.
It can be derived by analyzing SSR- Squared Sum of Residuals under Gauss-Markov Assumptions.

Gauss-Markov OLS assumptions:

1. **Linearity:** $y_i = \sum_{i=1}^{p} x_{ij}\,\beta_j + \epsilon_i \quad \forall i$

2. **Independence:** Observations are independent of each other. $\epsilon_i \perp \epsilon_j \;\; \forall i \neq j, \;\; \mathrm{Cov}(\epsilon_i, \epsilon_j) = 0 \;\; \forall i \neq j$

3. **Homoscedasticity:** $Var(\epsilon_i) = \sigma^2 \;\; \forall i$

4. **No Perfect Multicollinearity:** Independent variables are not perfectly correlated. This guarantees that $(X^{\mathrm{T}}X)$ **is invertible**. $rank(X) = p + 1$

5. **Zero Mean of Errors:** The error terms have an expected value of zero. $E[\epsilon_i] = 0$

6. **No Autocorrelation:** Error terms are uncorrelated across observations. $\mathrm{Cov}(\epsilon_i, \epsilon_j) = 0 \;\; \forall i \neq j$

7. **Normality (for inference):** Error terms are normally distributed (required for hypothesis testing and constructing confidence intervals). $\epsilon_i \sim \mathcal{N}(0, \sigma^2) \;\; \forall i$

when 1-7 are hold, then OLS is the **Best Linear Unbiased Estimator (BLUE)**

Maximum Likelihood Estimation (MLE)

Maximizes the likelihood that the model parameters would produce the observed data.

Assumptions: MLE assumptions (like Gauss-Markov assumptions)

Given ϵ_i are normally distributed and independent, the **likelihood function** $L(\beta, \sigma^2|y) =$ for the observed data $\mathrm{Y} = [y_1, y_2, \dots y_n]^T$ is defined as:

$$L(\beta, \sigma^2|y) = \prod_{i=1}^{n} f(y_i|x_i, \beta, \sigma^2)$$

,where $f(y_i|x_i, \beta, \sigma^2)$ is the probability density function of y_i .
Given the Normality assumption from G-M assumptions: $y_i|x_i, \beta, \sigma^2 \sim \mathcal{N}(x_i^T\beta, \sigma^2) \quad \forall i$

$$L(\beta, \sigma^2|y) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - x_i^T\beta)}{2\sigma^2}\right)$$

Log-Likelihood Function

Maximizing the **likelihood** is equivalent to maximizing the **log-likelihood.**

$$\ell(\beta, \sigma^2|y) = \ln L(\beta, \sigma^2|y)$$

$$= -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - x_i^T\beta)^2$$

<u>Maximizing the Log-Likelihood</u>

$$\frac{d\ell}{d\beta} = \frac{1}{\sigma^2}X^T(y - X\beta) = 0 \quad \longrightarrow \quad X^Ty = X^TX\beta \quad \longrightarrow \quad \boxed{\hat{\beta} = (X^TX)^{-1}X^Ty}$$

$$\frac{d\ell}{d\sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(y_i - x_i^T\beta)^2 = 0$$

$$\boxed{\widehat{\sigma^2} = \frac{1}{n}\sum_{i=1}^{n}(y_i - x_i^T\beta)^2 = \frac{1}{n}e^Te}$$

Biased estimation of sigma

Where $\mathrm{e} = y - X\hat{\beta}$ is the vector of residuals.

SSR: Squared Sum of Residuals $e^Te = SSR$

MLE Assumptions

- **Correct Model Specification:** The true data-generating process is correctly specified.
- **Identifiability:** The parameters are identifiable from the distribution.
- **Independence:** Observations are independent. $\epsilon_i \perp \epsilon_j \;\; \forall i \neq j$
- **Distributional Assumption:** Errors follow the specified distribution (e.g., normality). $\epsilon_i \sim \mathcal{N}(0, \sigma^2) \;\; \forall i$
- **Full Rank:** The predictor matrix X has full rank. $rank(X) = p + 1$
- **Continuity and Differentiability:** The likelihood function is sufficiently smooth for differentiation.

Note: While OLS does not inherently require normality for the estimation of β, MLE does require normality for its derivation in linear regression.
However, OLS still requires normality for exact inference (e.g., confidence intervals, hypothesis testing).

OLS VS MLE FOR LINEAR REGRESSION

OLS

$$\boxed{\hat{\beta} = (X^T X)^{-1} X^T y}$$

Target function

$$S(\beta) = (y - X\beta)^{\mathrm{T}}(y - X\beta) \rightarrow min$$

Under the Gauss-Markov assumptions, OLS estimators are **BLUE**—they have the smallest variance among all linear unbiased estimators.

$$E\left[\widehat{\sigma^2_{OLS}}\right] = \sigma^2$$

Unbiased estimation

MLE

$$\boxed{\hat{\beta} = (X^{\mathrm{T}} X)^{-1} X^{\mathrm{T}} y}$$

Target function

$$\ell(\beta, \sigma^2 | y) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}\left(y_i - x_i^T\beta\right)^2 \rightarrow max$$

Under regularity conditions, MLE estimators are **asymptotically efficient.** However, in finite samples, especially for σ^2, MLE may be less efficient due to bias.

$$E\left[\widehat{\sigma^2_{MLE}}\right] = \frac{n-p-1}{n}\sigma^2$$

Biased estimation

In large samples, the bias in σ^2 for MLE is so small, it makes both estimators comparable.

Most statistical packages default to OLS for variance estimation due to its unbiasedness, ensuring reliable inferential statistics.

BLUE: Best Linear Unbiased Estimator

Multicollinearity Problem

Multicollinearity occurs when **two or more independent variables are highly correlated**, leading to unstable coefficient estimates and inflated standard errors.

- Detection Techniques:
  - **Variance Inflation Factor (VIF):** Measures how much the variance of a regression coefficient is inflated due to multicollinearity.
  - **Correlation matrix analysis:** Helps detect pairs of features with high correlations, which indicate multicollinearity.
- Solutions:
  - **Dropping Correlated Features**
  - **Regularization Methods**: Ridge, Lasso, ElasticNet
  - **Principal Component Regression (PCR)**: Dimensionality reduction as an advanced solution to multicollinearity

Impact of Multicollinearity on Coefficient Estimation

$$\boxed{\hat{\beta} = (X^T X)^{-1} X^T y}$$

When there is multicollinearity:

1. $X^T X$ is nearly singular (determinant close to zero)
2. As a result of 1., $(\boldsymbol{X^T X})^{-1}$ becomes **unstable or large**, leading to **high variance in $\widehat{\boldsymbol{\beta}}$ coefficients**
3. Small changes in the data can lead to large fluctuations in $\hat{\beta}$, making the estimates **unreliable**

Variance Inflation Factor

$$\boxed{VIF(X_i) = \frac{1}{1 - R_i^2}}$$

where R_i^2 is the R^2 value obtained by regressing X_i on all other independent variables. **If $\boldsymbol{R^2}$ is close to 1, it means $\boldsymbol{X_i}$ is highly correlated with the other predictors**, leading to a large VIF value and indicating multicollinearity.
For i=1 we'll need to estimate R^2 for $X_1 = \beta_0 + \beta_2 X_2 + \cdots + \beta_p X_p$.

Rule of Thumb:
•VIF > 5 or VIF > 10 often signals significant multicollinearity

Basic Regularization Techniques

- Ridge,
- Lasso,
- Elastic Net

Ridge Regression (L2 Regularization)

Purpose: Penalizes large coefficients by adding an L2 penalty to the cost function. Reduces overfitting by **shrinking the coefficients towards zero without making them exactly zero.**

Loss Function:

$$\text{Minimize: } \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} \hat{\beta}_j^2$$

Where:

- $\boldsymbol{\lambda}$ **is the tuning parameter** controlling shrinkage (larger λ means more shrinkage).
- $\hat{\beta}_j^2$ are the regression coefficients.

Effect: Retains all features but reduces their impact.

Key Applications: many predictors with small to moderate effects and when multicollinearity is an issue.

Lasso Regression (L1 Regularization)

Purpose: Performs both shrinkage and feature selection by adding an L1 penalty to the cost function. Shrinks some coefficients to zero, effectively performing feature selection.

Loss Function:

Minimize: $\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} |\hat{\beta}_j|$

Where:

- $\boldsymbol{\lambda}$ **is the tuning parameter** controlling shrinkage (larger λ can set more coefficients to zero.).
- $\hat{\beta}_j^2$ are the regression coefficients.

Effect: Sets some coefficients to exactly zero, effectively removing less important features.

Key Applications: the model has many irrelevant features or automatic feature selection.

ElasticNet (Combined L1 and L2 Regularization)

Purpose: Combines the benefits of Ridge and Lasso by using both L1 and L2 regularization terms. Controls the complexity of the model by tuning the mix of both penalties.

Loss Function:

$$\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda_1 \sum_{j=1}^{p} |\hat{\beta}_j| + \lambda_2 \sum_{j=1}^{p} \hat{\beta}_j^2$$

Where:

- λ_1 controls the L1 penalty (Lasso-like behavior),
- λ_2 controls the L2 penalty (Ridge-like behavior).

Effect: Allows for a balance between Ridge and Lasso, achieving both feature selection and shrinkage.

Key Applications: Suitable when the number of predictors is much larger than the number of observations and when there is multicollinearity.

Comparison of Ridge, Lasso, and ElasticNet

Aspect	Ridge Regression	Lasso Regression	ElasticNet Regression
Penalty Type	L2	L1	L1 + L2
Feature Selection	No	Yes	Yes
Effect on Coefficients	Shrinks, no zeros	Shrinks, some zeros (removes features)	Shrinks, some zeros
Multicollinearity	Handles well	May struggle	Handles well
Use Case	When all features are important	When only some features are important	When balance between selection and shrinkage is needed
Tuning Parameters	1 parameter	1 parameter	2 parameters

Handling Data in Regression Models

- **Handling Missing Data:** Imputation (mean, median, mode).

- **Feature Scaling:** Important for regularized models (e.g., Ridge, Lasso).

- **Encoding Categorical Variables:**
  One-Hot Encoding, ← **Keep an eye on Multicollinearity!**
  Label Encoding for categorical features.

How to measure the results?

Metric	Description	Formula
Mean Squared Error (MSE)	Commonly used metric	$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$
Root Mean Squared Error (RMSE)	preferred when interpretability matters. The metric is in the same units as the target variable.	$RMSE = \sqrt{MSE}$
Mean Absolute Error	Robust against outliers.	$MAE = \frac{1}{n}\sum_{i=1}^{n}\lvert y_i - \hat{y}_i \rvert$
R-Squared $(\boldsymbol{R^2})$	Represents the proportion of variance explained by the model. Ranges from 0 to 1., where 1 indicates a perfect fit.	$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y}_i)^2}$
Adjusted $\boldsymbol{R^2}$	Adjusted for the number of predictors.	$R^2_{adj} = 1 - \frac{(1 - R^2)(n-1)}{n-p-1}$
Residuals Analysis	Involves checking the distribution of residuals to ensure randomness, as required by the assumptions of linear regression.	$SSR = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$

Practical Considerations

Data Splitting & Cross-Validation for Regression

- **Train-Test Split**: Split the dataset into training (for model training) and testing (for model evaluation) sets. Common splits: 70%-80% training, 20%-30% testing.

- **K-Fold Cross-Validation**: Divide data into k folds (commonly 5 or 10). Train on k−1k−1 folds, test on the remaining fold, repeat for all folds. Average results across folds for a more robust evaluation.

- **Stratified Cross-Validation:** For regression tasks, ensure each fold contains similar target value distributions (stratified on target percentiles).

Practical Considerations

Overfitting: High Training Accuracy + Low Test Accuracy.

Solutions:

- Regularization (Ridge, Lasso)
- Cross-validation
- Simplifying the model

Underfitting: Low Training and Test Accuracy

Solutions:

- Add complexity (e.g., more features or higher-order terms)
- Try non-linear models (e.g., polynomial regression)

Interpretability of Regression Models

Linear Regression:

- Coefficients directly explain how each feature impacts the target.
- Easy to interpret because it assumes a linear relationship between variables.

- **Complex Models** (e.g., Random Forest, Gradient Boosting):
  - Require feature importance methods like SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-Agnostic Explanations) to understand feature contributions.
  - Less interpretable but often more accurate for non-linear relationships.

Python Libraries for Regression

Scikit-Learn:

- **Linear Regression**, **Ridge**, **Lasso**, **Elastic Net**, **Polynomial Regression**
- **Random Forest Regressor**, **Support Vector Regression (SVR)**, and **Gradient Boosting**.
- Lacks flexibility for complex statistical analysis

Statsmodels:

- OLS, **Generalized Linear Models (GLMs)**, **Logit/Probit regression**, **Time series models**, and more.
- Detailed statistical summaries, including p-values, confidence intervals, and hypothesis testing.
- Provides rich model diagnostics (e.g., checking assumptions of linear regression).

TensorFlow/Keras/PyTorch:

- Represent DL task as regression: input – dense (features) – activation (relu/linear) – dense 1
- Ideal for complex, non-linear regression tasks that require deep learning models.

Advanced Regression Models

These advanced models go beyond simple linear regression and OLS:

- Logistic Regression
- Quantile Regression
- Polynomial Regression
- Poisson Regression
- Support Vector Regression (SVR)
- Generalized Linear Models (GLMs)

Each method offers unique advantages depending on the nature of the data and the prediction task at hand.

Live Coding & Demos

Summary and Wrap-Up

Key Takeaways & Final Thoughts

- **Continuous Targets:** Regression predicts continuous values (e.g., house prices, temperatures).
- **Model Evaluation:** Use metrics like **MSE, RMSE, MAE**, and **R-squared** to assess performance.
- **Multicollinearity:** Manage using Ridge, Lasso, and ElasticNet regularization.
- **Data Handling:** Properly scale features, handle missing data, and encode categorical variables.
- **Model Validation:** Apply cross-validation (e.g., K-Fold) to prevent overfitting.
- **Monitor Overfitting/Underfitting:** Regularization and model tuning are essential for optimal performance.

Thank you for your attention

Advanced Topics in Regression (OPTIONAL)

Logistic Regression

Logistic regression is often seen as a classification model but technically it's a regression

Logistic regression models the **probability of a binary outcome** (e.g., success/failure) based on independent variables. The output is constrained between 0 and 1 using the sigmoid function.

Formula:

$$log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

Where p is the probability of the positive class.

Applications: Predicting whether a customer will churn or not, predicting disease presence, etc.

https://scikit-learn.org/1.5/modules/generated/sklearn.linear_model.LogisticRegression.html

Quantile Regression

Quantile regression **estimates the conditional quantile** (like the median or the 90th percentile) of the dependent variable, rather than the mean, making it more robust to outliers.

Formula:

$$\min_{\beta} \sum_{i=1}^{n} \rho_\tau(y_i - X_i\beta)$$

Where ρ_τ is a quantile loss function for quantile τ.

Applications: Used in financial risk modeling (e.g., VaR - Value at Risk) and cases where you want to predict extremes.

https://scikit-learn.org/1.5/modules/generated/sklearn.linear_model.QuantileRegressor.html

Polynomial Regression

Polynomial regression **models non-linear relationships** by introducing polynomial terms (squared, cubed, etc.) of the features.

Formula: $$y = \beta_0 + \beta_1 x + \beta_2 x^2 \ldots + \beta_m x^m$$

Applications: Used in cases where the relationship between variables is curvilinear, such as modeling growth curves or trends in data that exhibit bends.

https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.PolynomialFeatures.html

Poisson Regression

Poisson regression is used when the **target variable represents count data** (e.g., the number of events happening in a fixed time interval).

Formula:

$$log(\lambda) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

Where λ is the expected count.

Applications: Used in insurance for predicting the number of claims, healthcare for the number of patients arriving at a clinic, etc.

OLS can't handle count data well due to non-constant variance, making Poisson regression a better alternative.

https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.PoissonRegressor.html

Support Vector Regression (SVR)

Support Vector Regression (SVR) attempts to fit the best line within a margin defined by the parameter ϵ, ignoring errors smaller than ϵ.

It tries to minimize a margin rather than minimizing squared error.

Formula: $\min_{\beta} \frac{1}{2} \|\beta\|^2$

Subject to the constraints $|y_i - X_i\beta| \leq \epsilon$

Applications: Used for complex non-linear regression tasks, such as stock market prediction or real estate pricing.

https://scikit-learn.org/stable/modules/generated/sklearn.svm.SVR.html

Generalized Linear Models (GLMs)

GLMs extend linear regression to handle non-normal error distributions (e.g., binomial, Poisson). They use a **link function** to relate the linear predictor to the mean of the distribution.

Formula: $g(\mu) = X\beta$

Where g is the link function, and μ is the expected value of the target variable.

Applications: Used when you need to model various distributions (e.g., logistic for binary outcomes, Poisson for counts).

Examples: Logistic regression is a GLM with a logit link function, and Poisson regression is a GLM with a log link function.

https://www.statsmodels.org/stable/glm.html